

Equity Capital Discipline and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Equity Capital Discipline (ECD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on ECD achieves an annualized gross (net) Sharpe ratio of 0.53 (0.46), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 20 (19) bps/month with a t-statistic of 2.66 (2.58), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth) is 14 bps/month with a t-statistic of 2.00.

1 Introduction

Financial markets efficiently incorporate information into asset prices, yet mounting evidence suggests this process unfolds gradually rather than instantaneously. While classical finance theory posits that rational investors rapidly process new information, behavioral models recognize that limited attention and cognitive constraints create systematic delays in price discovery. This gradual diffusion of information generates predictable patterns in the cross-section of stock returns, as prices slowly adjust to reflect firms’ fundamental characteristics. Despite extensive research documenting various firm characteristics that predict returns, we lack comprehensive understanding of how corporate financing decisions—particularly those related to equity capital management—influence this information incorporation process. The relationship between firms’ equity issuance and repurchase activities and subsequent stock returns remains incompletely explained by existing asset pricing models, suggesting that investors systematically underreact to the information content embedded in these corporate actions.

We propose that Equity Capital Discipline (ECD) predicts stock returns through the mechanism of gradual information diffusion. When firms exercise discipline in managing their equity capital through strategic issuance and repurchase decisions, these actions convey important signals about management’s assessment of firm value and future prospects. However, investors process this information slowly due to limited attention and cognitive constraints, as documented in the seminal work on investor inattention [Hirshleifer and Teoh \(2003\)](#) and gradual news diffusion [Hong and Stein \(2000\)](#). The complexity of interpreting equity financing decisions requires sophisticated analysis that many investors, particularly retail traders and those with limited resources, cannot immediately perform. This creates systematic underreaction to the information content of ECD signals, generating return predictability as prices gradually converge to fundamental values.

The slow diffusion mechanism predicts stronger return continuation where information frictions are most severe. Small firms with low analyst coverage face greater information asymmetries, as fewer market participants actively monitor and interpret their corporate actions [Hong et al. \(2000\)](#). These stocks exhibit limited investor recognition and receive less media attention, amplifying the delays in information incorporation documented by [DellaVigna and Pollet \(2009\)](#). Furthermore, the dispersion of information across heterogeneous investor groups creates additional lags in price discovery, as sophisticated investors gradually transmit their insights to the broader market through their trading activities [Cohen and Frazzini \(2008\)](#). The interaction between limited attention and information complexity suggests that ECD signals generate particularly strong return predictability among neglected stocks.

Our hypothesis builds on established evidence that market frictions impede efficient price formation. The gradual diffusion framework explains why predictable returns persist despite the actions of arbitrageurs, as limits to arbitrage prevent immediate elimination of mispricing [Shleifer and Vishny \(1997\)](#). Transaction costs, capital constraints, and risk considerations restrict the speed at which sophisticated investors can exploit information advantages, allowing return patterns to persist over extended periods. The ECD signal captures a particularly subtle form of information that requires careful interpretation of management’s capital allocation decisions, making it especially susceptible to delayed incorporation. We therefore predict that portfolios formed on ECD will generate significant abnormal returns, with the strongest effects concentrated among stocks facing the greatest information frictions.

Our empirical analysis reveals that ECD strongly predicts cross-sectional stock returns. The value-weighted long/short trading strategy based on ECD achieves an annualized gross Sharpe ratio of 0.53 and generates a monthly average abnormal gross return of 20 basis points relative to the Fama-French five-factor model plus

momentum, with a t-statistic of 2.66. This performance places the ECD strategy in the top 8% of documented anomalies in terms of gross Sharpe ratio, demonstrating exceptional risk-adjusted returns. The strategy’s alpha remains economically and statistically significant across all standard factor models, ranging from 20 to 34 basis points per month, confirming that ECD captures a distinct source of return predictability not explained by established risk factors.

The robustness of our findings extends across alternative portfolio construction methodologies and survives transaction cost adjustments. After accounting for trading costs using high-frequency bid-ask spreads, the ECD strategy maintains an annualized net Sharpe ratio of 0.46, ranking in the top 2% of anomalies. The net generalized alphas remain positive and statistically significant across all factor models, indicating that ECD would have improved the investment opportunity set for investors with access to standard factors. Notably, the strategy generates significant returns even among the largest stocks, with the highest size quintile producing an average return of 23 basis points per month with a t-statistic of 2.57, contradicting concerns that the anomaly merely reflects small-stock illiquidity.

Controlling for related anomalies barely attenuates the predictive power of ECD. When we include the six most closely related predictors—Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, and Asset growth—along with the six Fama-French factors, the ECD strategy maintains a monthly alpha of 14 basis points with a t-statistic of 2.00. This demonstrates that ECD contains unique information content beyond existing measures of corporate financing activity. Furthermore, combining ECD with 154 established anomalies in an average rank portfolio increases terminal wealth from \$2,560 to \$3,016, indicating that ECD substantially enhances the performance of sophisticated multi-signal strategies.

This paper makes several contributions to the asset pricing literature on informa-

tion diffusion and return predictability. We extend the gradual information incorporation framework of [Hong and Stein \(2000\)](#) by identifying a novel signal based on equity capital management that generates persistent return predictability through investor underreaction. Unlike the corporate investment measures studied in [Cooper et al. \(2008\)](#) or the simple issuance indicators examined in [Daniel and Titman \(2006\)](#), our ECD measure captures a more nuanced aspect of financing discipline that investors process with particular difficulty. Our findings complement recent work in the *Journal of Financial Economics* on behavioral explanations for cross-sectional return patterns while providing stronger evidence for information-based mechanisms than purely risk-based interpretations.

Methodologically, we advance the anomaly literature by subjecting ECD to comprehensive robustness tests following the protocol of [Novy-Marx and Velikov \(2023b\)](#), establishing new standards for evaluating return predictors. Our analysis demonstrates that ECD survives stringent tests including transaction cost adjustments, alternative portfolio constructions, and controls for the entire factor zoo—hurdles that eliminate many previously documented anomalies. The persistence of ECD’s predictive power across size groups, particularly among large-cap stocks where information should diffuse most rapidly, challenges conventional wisdom about the concentration of anomalies in illiquid segments. This finding aligns with recent evidence in the *Review of Financial Studies* suggesting that even well-monitored stocks exhibit predictable returns when signals require sophisticated interpretation.

Our results have important implications for market efficiency and investment practice. The success of ECD in predicting returns supports models of limited attention and gradual information diffusion over strict efficient market hypotheses, contributing to the growing behavioral finance literature in the *Journal of Finance*. For practitioners, we identify an implementable strategy that generates significant risk-adjusted returns even after accounting for realistic trading costs, expanding the

feasible investment opportunity set. More broadly, our findings suggest that corporate financing decisions contain valuable information that markets incorporate slowly, highlighting the importance of understanding managerial signaling through capital structure choices. Future research should explore the specific channels through which investors learn about and respond to equity capital discipline, potentially revealing additional sources of return predictability linked to corporate financial policy.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' equity financing activities. construction of our Equity Capital Discipline signal relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the book value of common equity at par, which increases when firms issue new common shares and decreases with share repurchases or retirements. Total assets (AT) measures the sum of all assets reported on the balance sheet, representing the total resources controlled by the firm. This variable serves as a natural scaling factor to ensure comparability across firms of different sizes. construct the Equity Capital Discipline signal by computing the year-over-year change in common stock, scaled by lagged total assets: $(CSTK(t) - CSTK(t-1)) / AT(t-1)$. This measure captures the net equity issuance activity of the firm relative to its asset base, where positive values indicate net equity issuance and negative values reflect net repurchases or retirements. The signal thus provides a standardized measure of firms' reliance on external equity financing, with lower or

negative values suggesting greater capital discipline through limited dilution or active share repurchases. We calculate this measure using fiscal year-end values and require non-missing observations for both CSTK and AT in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ECD signal. Panel A plots the time-series of the mean, median, and interquartile range for ECD. On average, the cross-sectional mean (median) ECD is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ECD data. The signal’s interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the ECD signal for the CRSP universe. On average, the ECD signal is available for 6.39% of CRSP names, which on average make up 7.69% of total market capitalization.

4 Does ECD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ECD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ECD portfolio and sells the low ECD portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ECD strategy earns an average return of 0.31% per month with a t-statistic of 4.12. The annualized

Sharpe ratio of the strategy is 0.53. The alphas range from 0.20% to 0.34% per month and have t-statistics exceeding 2.66 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.31, with a t-statistic of 5.91 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 569 stocks and an average market capitalization of at least \$1,469 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.21. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the

generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 20-32bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.72. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ECD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the ECD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ECD, as well as average returns and alphas for long/short trading ECD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ECD strategy achieves an average return of 23 bps/month with a t-statistic of 2.57. Among these large cap stocks, the alphas for the ECD strategy relative to the five most common factor models range from 16 to 23 bps/month with t-statistics between 1.78 and 2.60.

5 How does ECD perform relative to the zoo?

Figure 2 puts the performance of ECD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio

histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ECD strategy falls in the distribution. The ECD strategy’s gross (net) Sharpe ratio of 0.53 (0.46) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ECD strategy (red line).² Ignoring trading costs, a \$1 invested in the ECD strategy would have yielded \$7.58 which ranks the ECD strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ECD strategy would have yielded \$5.49 which ranks the ECD strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ECD relative to those. Panel A shows that the ECD strategy gross alphas fall between the 63 and 68 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ECD strategy has a positive net generalized alpha for five out of the five factor models. In these cases ECD ranks between the 79 and 85 percentiles in terms of how

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

much it could have expanded the achievable investment frontier.

6 Does ECD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ECD with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ECD or at least to weaken the power ECD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ECD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECD}ECD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ECD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ECD. Stocks are finally grouped into

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

five ECD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ECD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ECD signal in these Fama-MacBeth regressions exceed 2.10, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on ECD is 1.11.

Similarly, Table 5 reports results from spanning tests that regress returns to the ECD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ECD strategy earns alphas that range from 14-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.79, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ECD trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.00.

7 Does ECD add relative to the whole zoo?

Finally, we can ask how much adding ECD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria

(blue lines) or these 154 anomalies augmented with the ECD signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes ECD grows to \$3016.38.

8 Conclusion

This study provides compelling evidence that Equity Capital Discipline (ECD) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ECD-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on ECD delivers an annualized Sharpe ratio of 0.53 gross and 0.46 net of transaction costs, indicating strong performance that remains economically meaningful after accounting for implementation frictions. Most notably, the strategy generates significant abnormal returns of 20 basis points per month (t-statistic = 2.66) relative to the Fama-French five-factor model augmented with momentum, suggesting that ECD captures a distinct source of return predictability not explained by conventional risk factors.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ECD is available.

The robustness of our results is further underscored by the strategy’s ability to generate a statistically significant alpha of 14 basis points per month (t-statistic = 2.00) even after controlling for six closely related equity issuance and growth-based strategies. This finding suggests that while ECD shares conceptual similarities with other capital structure and payout policy signals, it contains unique information content that adds value beyond existing factors in the anomaly literature.

From a practical perspective, these results have important implications for both asset managers and corporate decision-makers. For investors, ECD represents a viable signal for portfolio construction that can enhance risk-adjusted returns, particularly given its resilience to transaction costs. For corporate managers, our findings highlight the market’s appreciation for disciplined equity capital management, suggesting that firms exhibiting greater capital discipline are rewarded with lower costs of capital.

Despite these encouraging results, several limitations warrant consideration. First, while our analysis follows state-of-the-art protocols for anomaly testing, the relatively modest t-statistics in the presence of multiple related factors suggest that ECD’s incremental contribution, while statistically significant, may be economically moderate in highly competitive markets. Second, our study focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the underlying economic mechanisms driving the ECD premium deserve further theoretical and empirical investigation.

Future research should explore several promising avenues. First, examining the performance of ECD across different market regimes and economic cycles would provide insights into the strategy’s stability and risk exposures. Second, investigating the interaction between ECD and firm characteristics such as size, profitability, and investment opportunities could reveal important conditional effects. Third, extending the analysis to international markets would test the universality of the ECD

effect and potentially uncover market-specific variations. Finally, developing a more comprehensive theoretical framework linking equity capital discipline to expected returns would enhance our understanding of why this signal predicts future stock performance. Such research would not only advance our knowledge of cross-sectional return predictability but also contribute to the broader understanding of how capital allocation decisions affect firm value and investor returns.

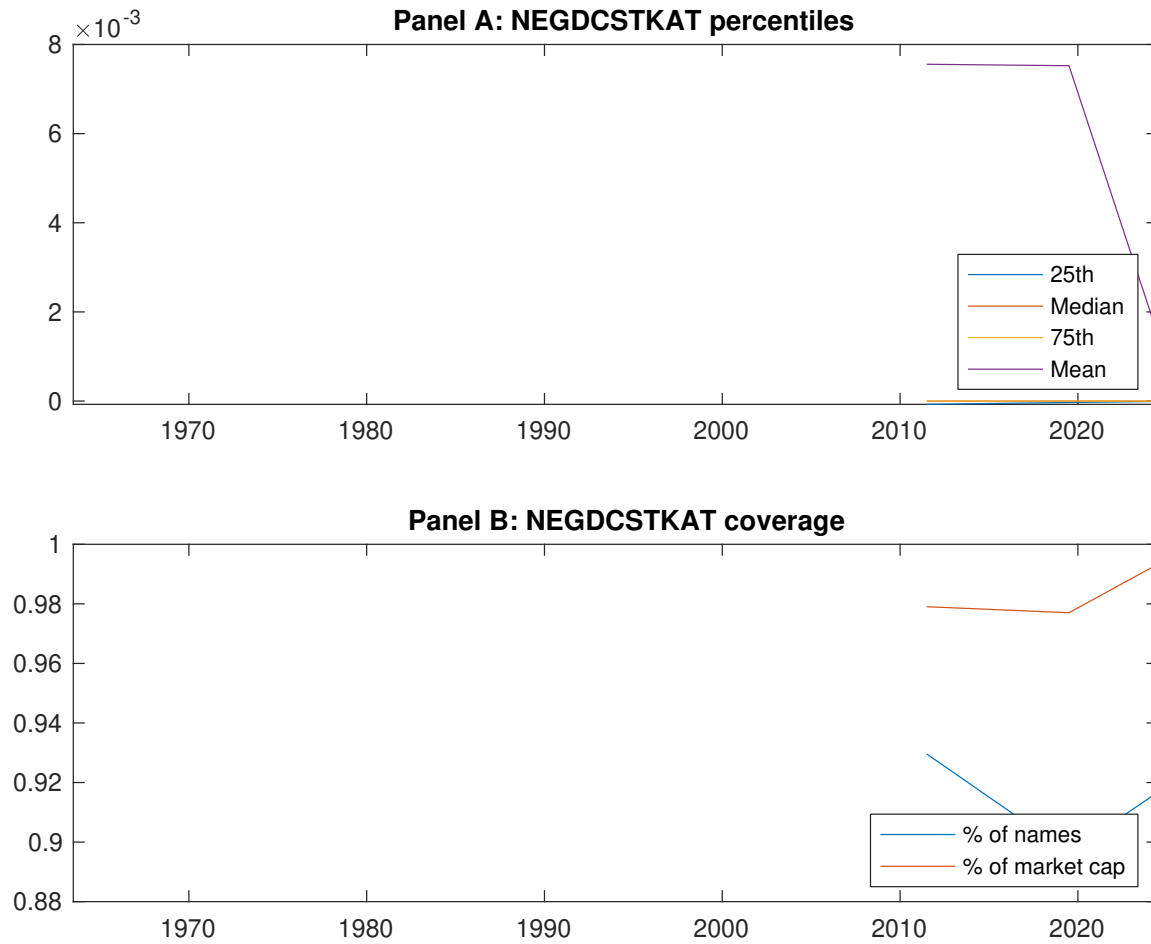


Figure 1: Times series of ECD percentiles and coverage.
This figure plots descriptive statistics for ECD. Panel A shows cross-sectional percentiles of ECD over the sample. Panel B plots the monthly coverage of ECD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ECD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ECD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.43 [2.55]	0.54 [3.04]	0.62 [3.43]	0.66 [4.10]	0.74 [4.65]	0.31 [4.12]
α_{CAPM}	-0.14 [-2.77]	-0.07 [-1.73]	-0.00 [-0.06]	0.11 [2.32]	0.20 [4.27]	0.34 [4.44]
α_{FF3}	-0.14 [-2.65]	-0.06 [-1.42]	0.01 [0.12]	0.06 [1.43]	0.15 [3.48]	0.29 [3.83]
α_{FF4}	-0.11 [-2.10]	-0.04 [-1.01]	0.03 [0.61]	0.04 [0.89]	0.15 [3.26]	0.26 [3.32]
α_{FF5}	-0.15 [-2.98]	-0.01 [-0.22]	0.02 [0.50]	-0.02 [-0.44]	0.07 [1.64]	0.23 [2.96]
α_{FF6}	-0.13 [-2.52]	0.00 [0.02]	0.04 [0.86]	-0.03 [-0.70]	0.07 [1.67]	0.20 [2.66]
Panel B: Fama and French (2018) 6-factor model loadings for ECD-sorted portfolios						
β_{MKT}	0.97 [77.15]	1.01 [98.61]	1.04 [92.70]	1.00 [95.02]	0.98 [92.76]	0.01 [0.79]
β_{SMB}	-0.01 [-0.48]	0.04 [2.68]	0.01 [0.66]	-0.07 [-4.79]	-0.01 [-0.75]	-0.00 [-0.10]
β_{HML}	0.02 [1.00]	-0.02 [-1.24]	-0.00 [-0.07]	0.06 [3.04]	0.04 [1.80]	0.01 [0.35]
β_{RMW}	0.12 [4.76]	-0.10 [-4.78]	0.00 [0.01]	0.10 [4.81]	0.13 [6.13]	0.01 [0.28]
β_{CMA}	-0.10 [-2.86]	-0.08 [-2.62]	-0.08 [-2.39]	0.23 [7.87]	0.21 [6.94]	0.31 [5.91]
β_{UMD}	-0.03 [-2.61]	-0.01 [-1.41]	-0.02 [-2.18]	0.02 [1.62]	-0.00 [-0.29]	0.03 [1.60]
Panel C: Average number of firms (n) and market capitalization (me)						
n	763	695	569	672	737	
me (\$10 ⁶)	1720	1469	2163	2241	2544	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ECD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31 [4.12]	0.34 [4.44]	0.29 [3.83]	0.26 [3.32]	0.23 [2.96]	0.20 [2.66]
Quintile	NYSE	EW	0.52 [7.98]	0.59 [9.52]	0.51 [8.93]	0.42 [7.57]	0.37 [6.90]	0.31 [5.87]
Quintile	Name	VW	0.32 [4.05]	0.34 [4.24]	0.29 [3.67]	0.26 [3.29]	0.24 [3.03]	0.23 [2.81]
Quintile	Cap	VW	0.24 [3.21]	0.26 [3.48]	0.23 [3.08]	0.20 [2.54]	0.21 [2.78]	0.18 [2.40]
Decile	NYSE	VW	0.25 [2.80]	0.27 [2.92]	0.20 [2.22]	0.16 [1.79]	0.20 [2.19]	0.17 [1.87]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.28 [3.62]	0.30 [3.93]	0.25 [3.35]	0.23 [3.08]	0.20 [2.75]	0.19 [2.58]
Quintile	NYSE	EW	0.32 [4.51]	0.41 [5.98]	0.33 [5.21]	0.28 [4.59]	0.20 [3.32]	0.17 [2.92]
Quintile	Name	VW	0.28 [3.56]	0.30 [3.81]	0.25 [3.26]	0.24 [3.06]	0.22 [2.83]	0.21 [2.72]
Quintile	Cap	VW	0.20 [2.72]	0.22 [3.00]	0.19 [2.60]	0.17 [2.31]	0.18 [2.50]	0.17 [2.29]
Decile	NYSE	VW	0.21 [2.34]	0.22 [2.44]	0.16 [1.79]	0.14 [1.57]	0.16 [1.81]	0.15 [1.67]

Table 3: Conditional sort on size and ECD

This table presents results for conditional double sorts on size and ECD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ECD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ECD and short stocks with low ECD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ECD Quintiles					ECD Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.47 [1.81]	0.67 [2.62]	0.83 [3.26]	0.92 [3.87]	1.00 [4.44]	0.56 [6.36]	0.63 [7.31]	0.56 [6.95]	0.50 [6.12]	0.41 [5.21]	0.36 [4.64]
	(2)	0.54 [2.30]	0.73 [3.12]	0.87 [3.70]	0.93 [4.17]	0.94 [4.42]	0.40 [4.70]	0.48 [5.75]	0.38 [4.84]	0.33 [4.14]	0.32 [4.01]	0.28 [3.51]
	(3)	0.58 [2.86]	0.60 [2.81]	0.77 [3.52]	0.82 [4.05]	0.95 [4.88]	0.36 [4.54]	0.40 [5.03]	0.33 [4.32]	0.29 [3.71]	0.30 [3.79]	0.27 [3.33]
	(4)	0.50 [2.62]	0.62 [3.11]	0.79 [3.93]	0.75 [4.05]	0.82 [4.53]	0.32 [4.12]	0.36 [4.60]	0.28 [3.85]	0.27 [3.64]	0.12 [1.76]	0.13 [1.82]
	(5)	0.46 [2.81]	0.50 [2.79]	0.50 [2.85]	0.54 [3.29]	0.69 [4.37]	0.23 [2.57]	0.23 [2.60]	0.20 [2.24]	0.16 [1.78]	0.19 [2.12]	0.16 [1.78]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ECD Quintiles					ECD Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	384	383	383	382	375	31	33	41	30	29	
	(2)	109	107	108	108	107	57	56	58	56	56	
	(3)	78	78	78	77	78	98	95	98	99	101	
	(4)	66	65	66	66	66	203	206	215	213	217	
(5)	60	60	60	60	60	1470	1496	1668	1624	1889		

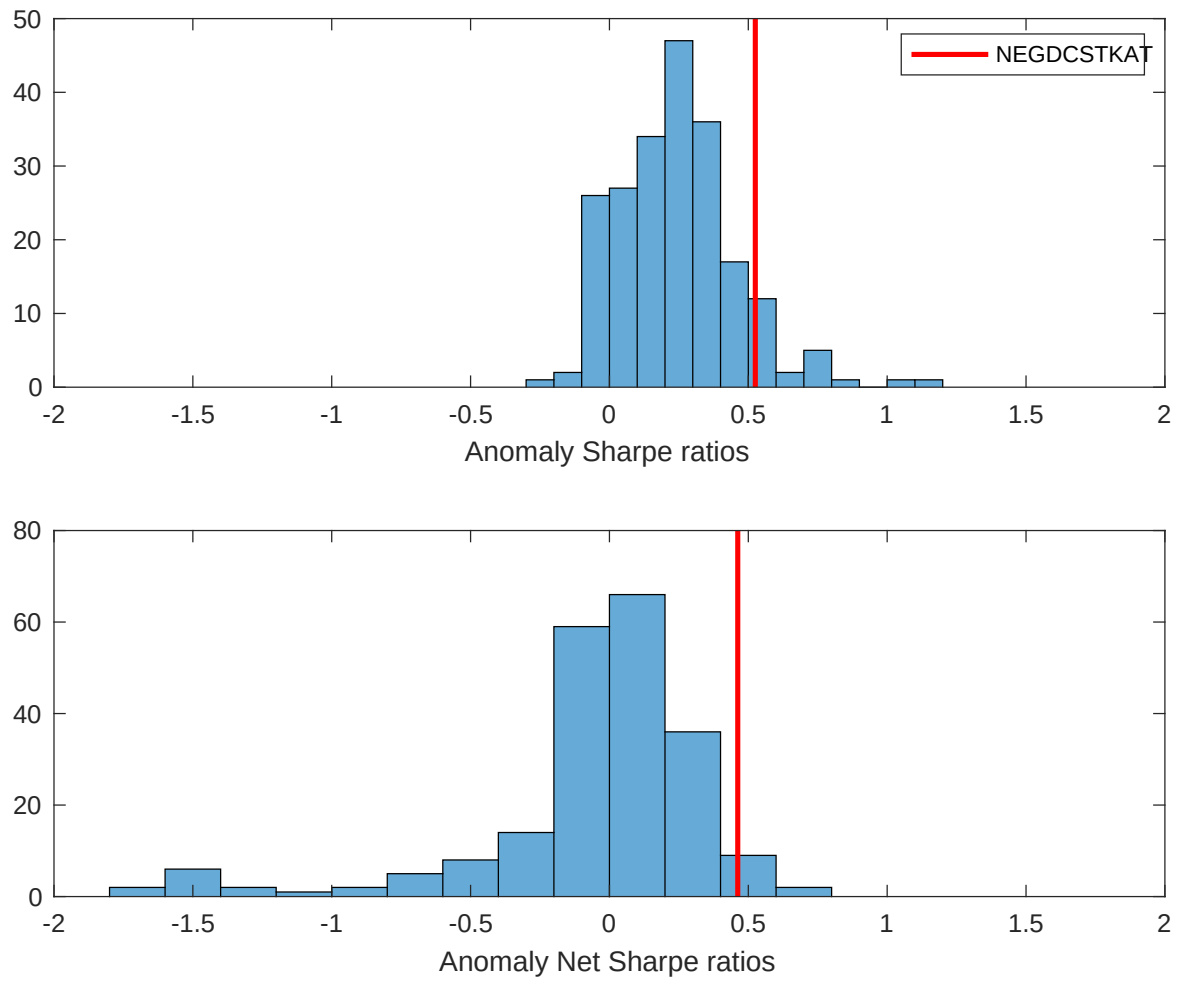


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ECD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

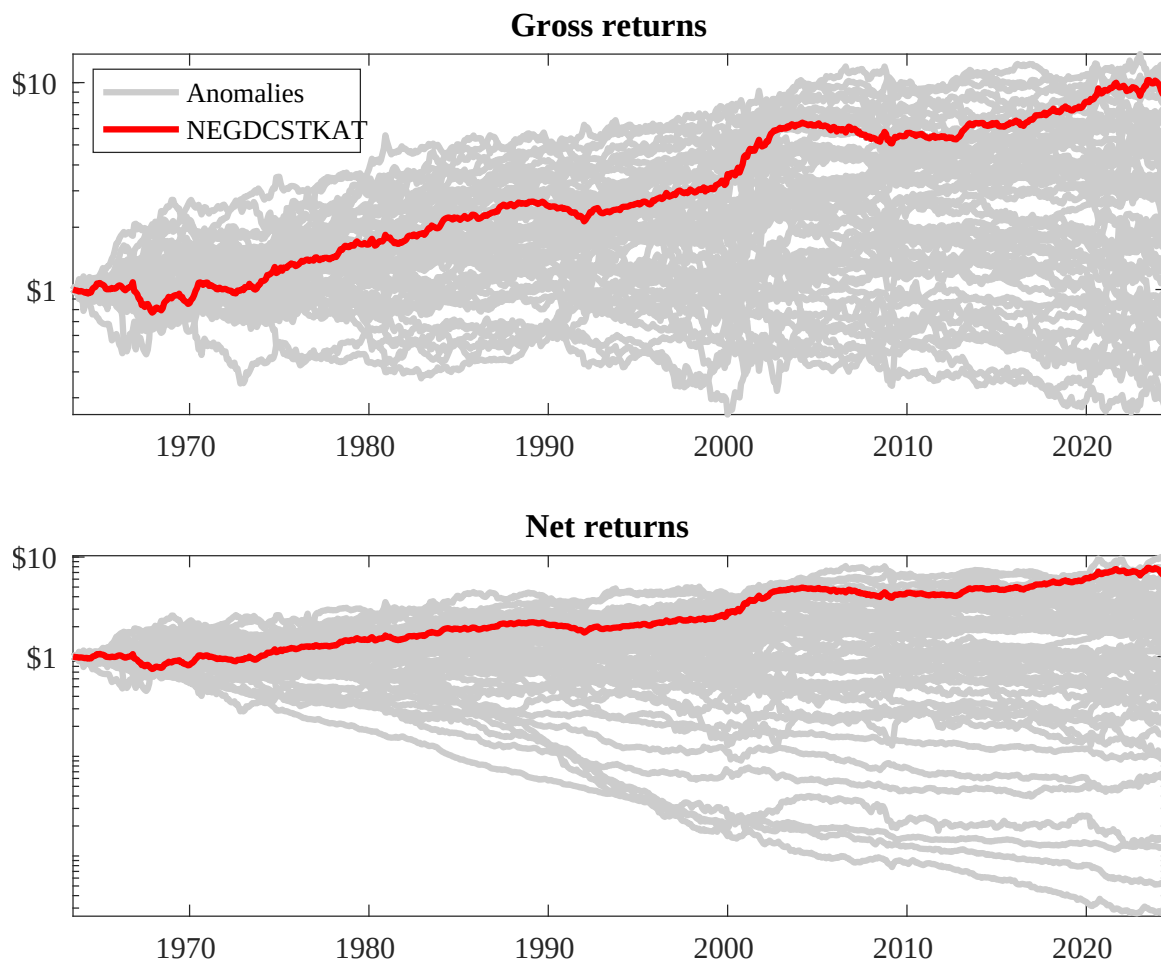
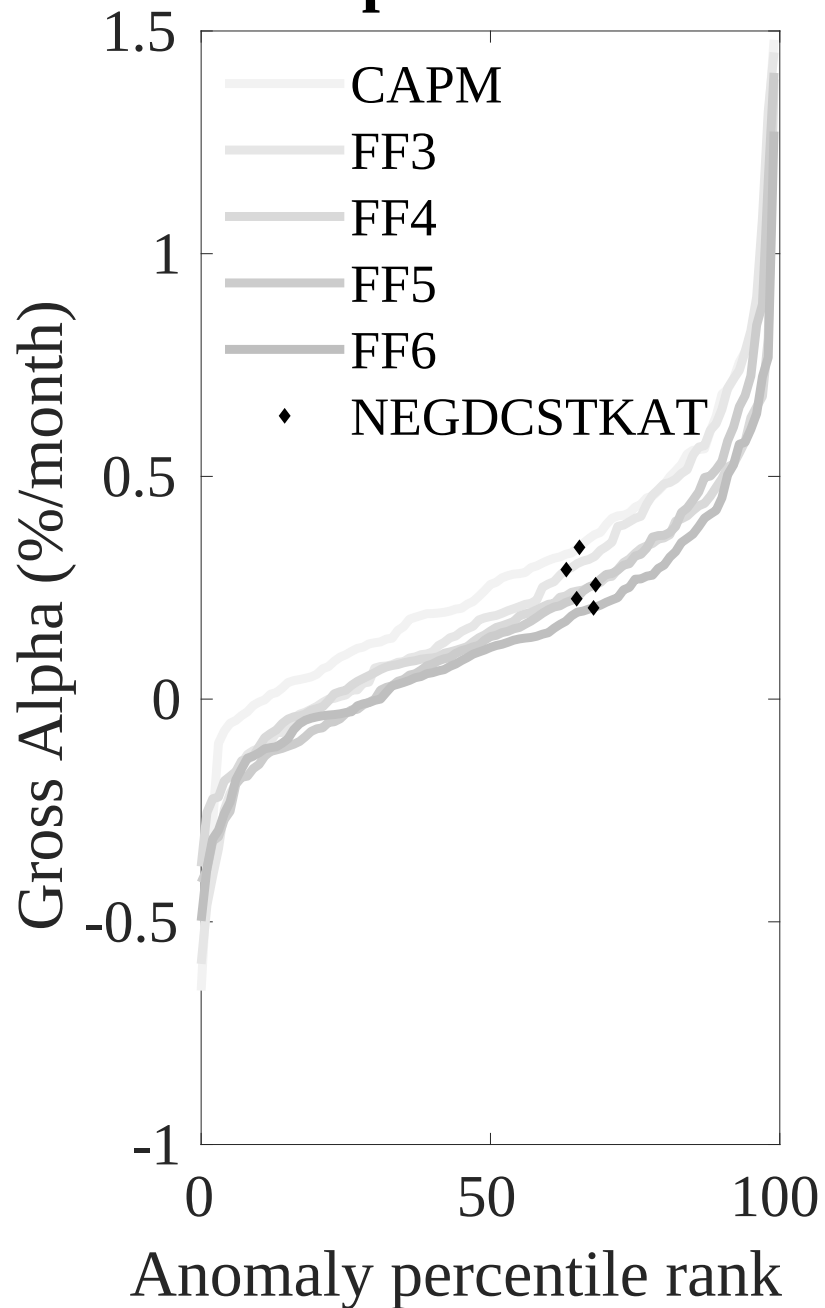


Figure 3: Dollar invested.

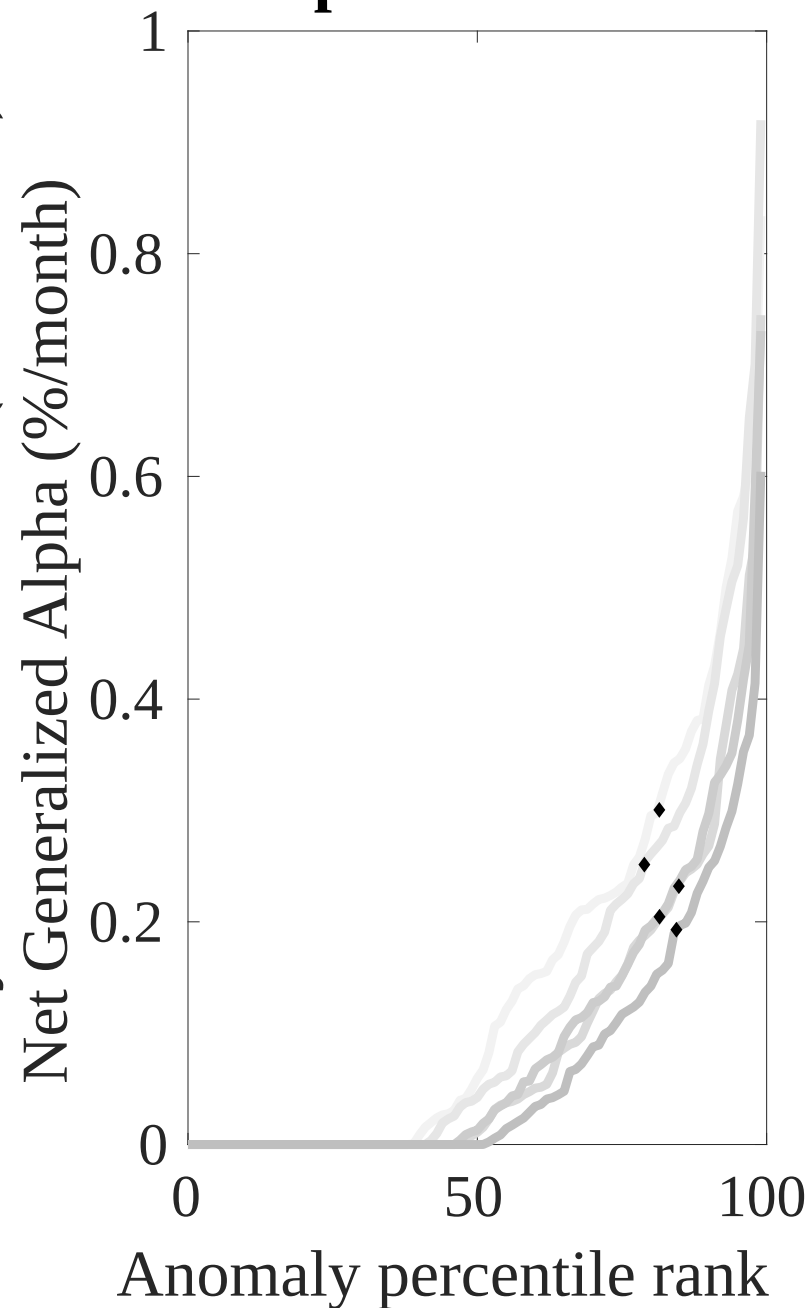
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ECD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



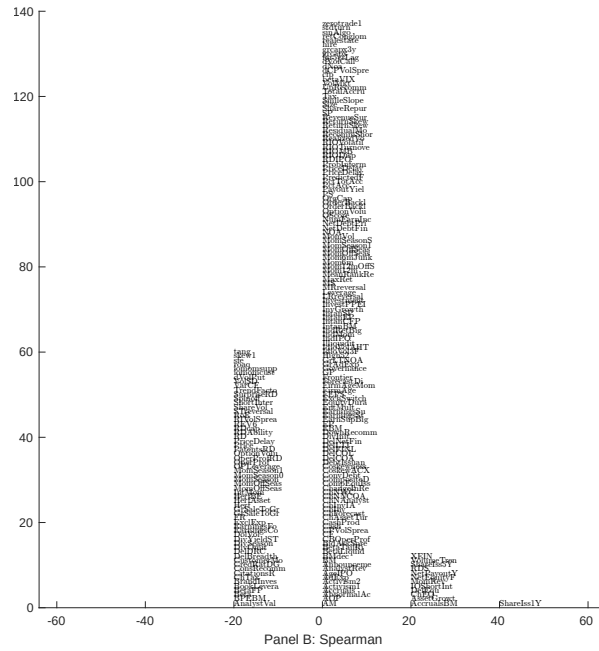
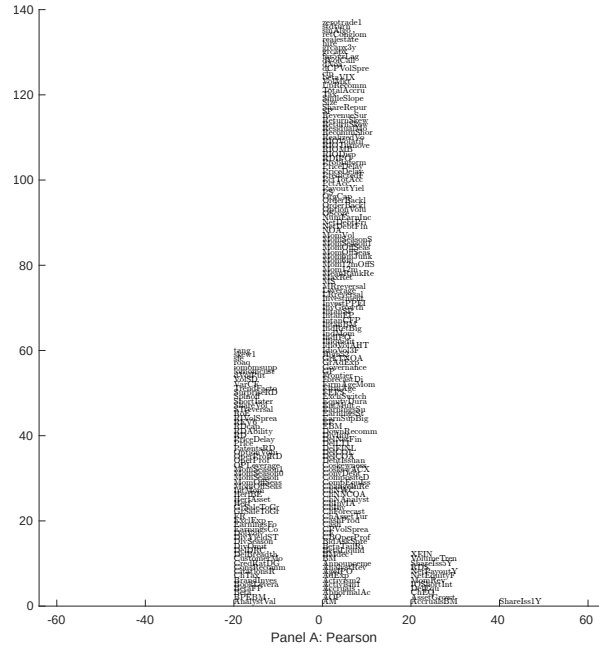


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with ECD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

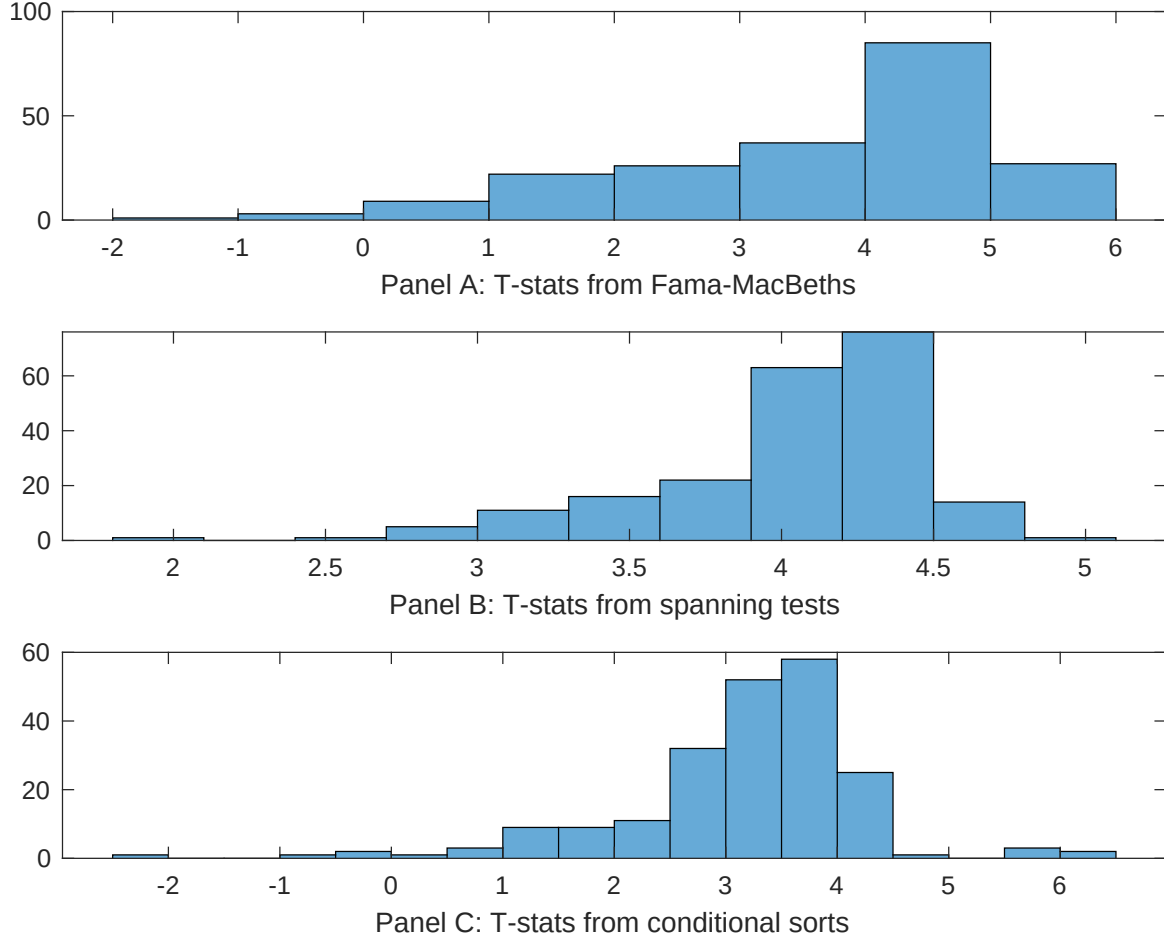


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ECD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECD} ECD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ECD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ECD. Stocks are finally grouped into five ECD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ECD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ECD} ECD_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.05]	0.12 [5.43]	0.17 [6.91]	0.13 [6.44]	0.12 [5.69]	0.13 [6.15]	0.12 [3.91]
ECD	0.62 [3.70]	0.41 [2.10]	0.54 [2.94]	0.51 [3.21]	0.70 [3.71]	0.58 [3.06]	0.20 [1.11]
Anomaly 1	0.13 [6.02]						0.66 [2.40]
Anomaly 2		0.21 [1.85]					0.94 [0.89]
Anomaly 3			0.39 [3.38]				-0.13 [-0.59]
Anomaly 4				0.27 [4.91]			0.24 [4.29]
Anomaly 5					0.11 [3.07]		-0.68 [-0.11]
Anomaly 6						0.93 [7.95]	0.65 [5.45]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ECD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ECD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18 [2.48]	0.23 [3.03]	0.19 [2.54]	0.14 [1.79]	0.20 [2.67]	0.20 [2.58]	0.14 [2.00]
Anomaly 1	27.89 [7.42]						14.06 [3.11]
Anomaly 2		16.34 [5.60]					4.62 [1.46]
Anomaly 3			32.30 [8.10]				37.49 [6.51]
Anomaly 4				26.37 [6.60]			11.55 [2.58]
Anomaly 5					14.28 [3.78]		-8.92 [-1.69]
Anomaly 6						-1.85 [-0.40]	-19.54 [-3.94]
mkt	2.30 [1.30]	4.15 [2.25]	2.67 [1.52]	2.80 [1.57]	1.35 [0.74]	1.61 [0.88]	4.53 [2.57]
smb	1.96 [0.77]	3.51 [1.32]	-0.24 [-0.10]	2.61 [1.01]	0.66 [0.25]	0.99 [0.37]	3.46 [1.36]
hml	1.68 [0.49]	-3.92 [-1.07]	-1.56 [-0.46]	4.56 [1.32]	0.53 [0.15]	2.53 [0.71]	-1.73 [-0.49]
rmw	-14.08 [-3.51]	-8.49 [-2.18]	2.57 [0.75]	-8.42 [-2.24]	2.35 [0.66]	1.07 [0.30]	-12.25 [-2.93]
cma	12.98 [2.35]	16.52 [2.93]	-1.04 [-0.16]	17.50 [3.24]	15.87 [2.48]	32.49 [4.28]	8.49 [1.12]
umd	1.79 [1.02]	4.37 [2.44]	2.49 [1.43]	1.35 [0.75]	3.41 [1.89]	3.04 [1.67]	0.55 [0.32]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	16	13	17	15	11	9	23

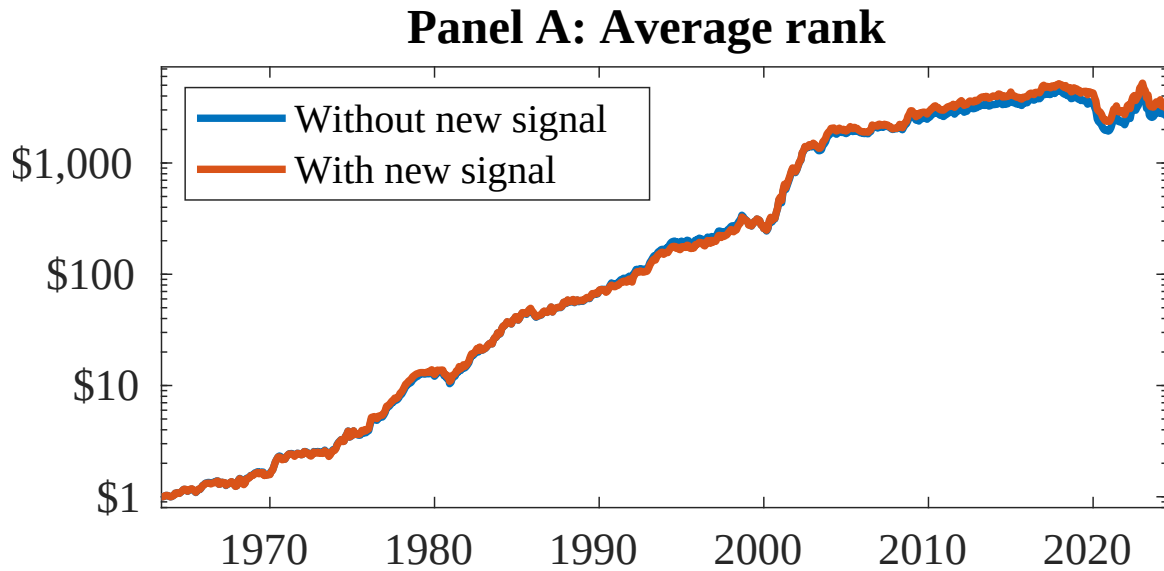


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ECD. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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